

Regional variations and trends in the use of maintenance inhaled medications in chronic obstructive pulmonary disease



Jeenat Mehareen¹, Yiwei Yin¹, Kayly Choy¹, Kate M. Johnson^{1,3,4}, Don D. Sin^{1,2}, Mohsen Sadatsafavi^{1,4}

1.Respiratory Evaluation Sciences Program, Faculty of Pharmaceutical Sciences, University of British Columbia, Vancouver, Canada;; 2.Centre for Heart Lung Innovation and Division of Respiriology, Department of Medicine, University of British Columbia, Vancouver,, Canada;; 3.Division of Respiratory Medicine, Department of Medicine, University of British Columbia, Vancouver, Canada; 4.Legacy for Airway Health, Vancouver Coastal Health Research Institute, Vancouver, Canada

BACKGROUND

Maintenance inhaled therapies form the foundation of COPD management.

Despite evolving treatment guidelines, little is known about how regional prescribing patterns shift over time, specially within the healthcare system.

This highlights the need to study temporal and geographical trends.

OBJECTIVES

To describe temporal and geographical trends in long-acting maintenance inhaled medication use within a single publicly funded healthcare system.

METHODS

- Design**
 - Retrospective Longitudinal population - based cohort study
- Data**
 - Population Data BC (Vital Statistics, Consolidation files, Medical Services Plan)
- Time**
 - 2010-2020
- Variable of Interest**
 - Health Services Delivery Area
- Outcome**
 - Proportion of COPD-related maintenance inhaled medication users among prevalent COPD patients, overall and by drug class
- Statistical Analysis**
 - Descriptive Statistics
 - Adjusted Generalized Linear Model
 - P<0.05 (Significance level)

RESULTS

Patients
189,204

Mean Age
68.0 years

Female
50.9%

Area of Residence
84% urban

Single Inhaler Therapies

ICS (33.4%) LAMA (19%) LABA (2.3%)

Combination Therapies

ICS+LABA (39.1) ICS+LAMA+LABA (14.7%) LAMA+LABA (6.4%)

Significant regional variability in medication use persists even after adjusting for patient factors.

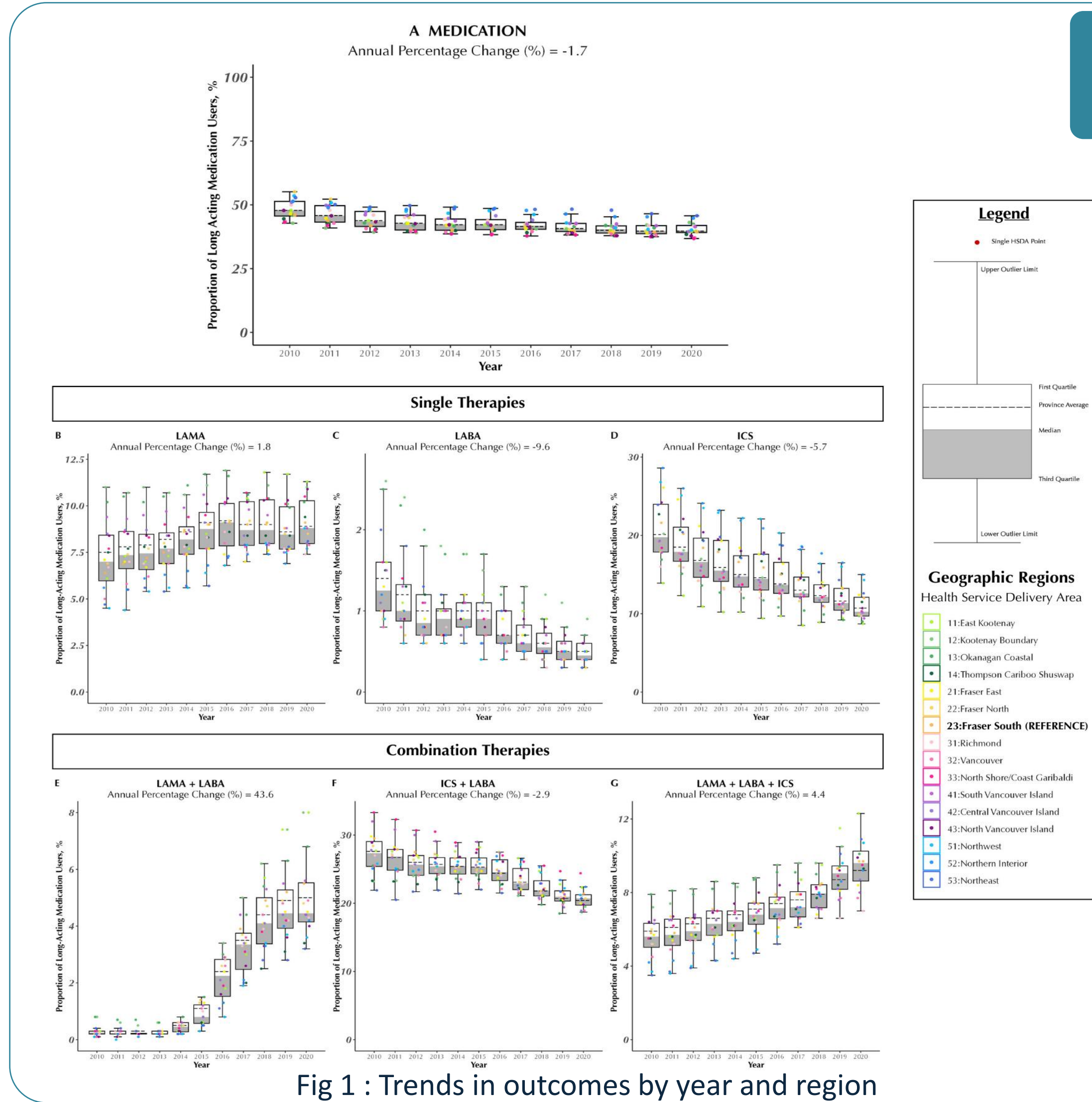


Fig 1 : Trends in outcomes by year and region

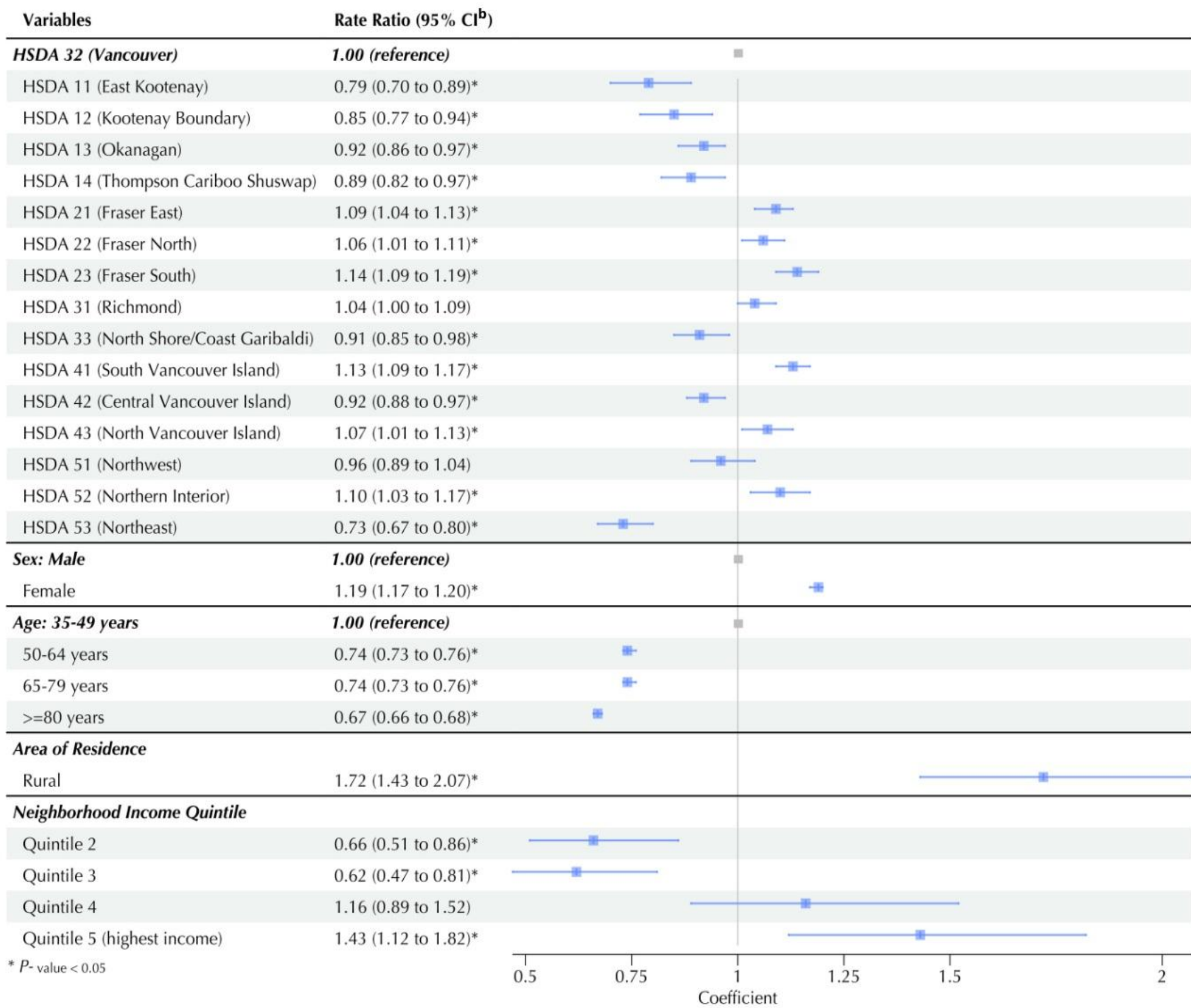


Fig 2: Rate Ratios in a forest plot

Our study showed an **overall decline in the proportion COPD patients using maintenance inhaled medications**, primarily driven by reduced proportion of ICS and/or LABA users

Despite the decrease **ICS-containing regimens** remained the most used medication across all years.

These findings highlight opportunities for **improving prescribing practices** and reducing unwarranted variation in COPD pharmacotherapy